

# Trent Wiles

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## EDUCATION

### Northeastern University — Khoury College of Computer Sciences

Boston, MA

*Candidate for Bachelor of Science in Computer Science and Criminal Justice*

*August 2023 — Present*

- GPA: 3.7/4.0; Dean's List for all semesters; expected graduation: May 2027
- Clubs & Organizations: ACM Chapter, Content Club, OSCRR Student Board Member
- Relevant Coursework: Object-Oriented Programming, Database Design, Software Development, Algorithms & Data Structures, Introduction to Math Reasoning, Discrete Math, Networking Fundamentals

## EXPERIENCE

### Amazon

June 2026 - August 2026

*Security & Loss Prevention Intern*

- Designed and helped implement a security screening plan at one of Amazon's top 3 highest-throughput fulfillment centers, screening the items of all 5,000 site associates to reduce insider theft, involving up to 800 associates at a time during peak hours
- Trained on standard loss prevention tools, discovering three theft cases valued at \$1,500 in three weeks
- Developed multiple iterations of a plexiglass conveyor belt prototype for associate possession screening, then authored a complete build guide with safety information, photos, and step-by-step instructions

### HITE Hedge Asset Management LLC

January 2026 - June 2026

*Software Engineer Co-op*

- Built a React and TypeScript AI chatbot UI under senior supervision that gives portfolio managers natural language access to commodities and automotive data from internal S3 and Databricks sources, surfacing company metrics to support investment decisions
- Deployed the chatbot to the company's production web infrastructure on EC2 and hardened it with Microsoft OAuth and firewall-level IP whitelisting to block unauthorized external access and maintain regulatory compliance
- Eliminated the repetitive cycle of downloading data, editing in Excel, and re-feeding it to the model by embedding a live-syncing CSV editor with formula support directly into the chat interface

### Kaleidoscope at Khoury College

September 2025 - Present

*Co-President*

- Will co-lead a student organization of 10 members alongside a fellow Co-President, coordinating events and initiatives with a faculty advisor to support Khoury College's club ecosystem
- Led the software team to manage and investigate technical resources for Northeastern Kaleidoscope, a club council overseeing 50+ student organizations funded by Khoury College
- Partnered with the operations team to automate Slack notifications integrated with Notion, streamlining communication across 10,000+ dollars in annual club distributions

## PROJECTS

### HackerNews Clone | Go, React, TypeScript, PostgreSQL, Docker

July 2025

- Built a production-grade HackerNews clone with magic link authentication, submission/comment posting with reCAPTCHA v3 protection, and community moderation flagging, deployed via Docker and Caddy
- Implemented full voting and sorting systems, user customization features, and comprehensive HTTP/business logic logging to enable 2,000+ monthly active users
- Engineered a Go backend and React/TypeScript frontend using ShadCN components, achieving sub-100ms API response times and support for 500+ users in production

### RAG Pipeline Metrics | Python, gRPC, PostgreSQL, Clickhouse, Cassandra, OpenSearch

June 2025

- Partnered with NetApp Instaclustr in a team of 8 during Northeastern's Software Engineering in Australia program to design a RAG pipeline using Mistral 7B that benchmarked query latency across four distributed database backends
- Developed scalable FastAPI microservices for document ingestion and embedding generation, leveraging gRPC for modular inter-service communication in a fast-paced Agile environment
- Optimized document processing pipeline using SPECTER2 embeddings to achieve 10,000 documents per hour throughput, demonstrating production-grade scalability for scientific PDF ingestion

## TECHNICAL SKILLS

**Languages:** Java, Python, JavaScript, TypeScript, HTML/CSS, Go, C, Kotlin

**Frameworks:** Node.js, Flask, ExpressJS, React, WSGI, JUnit, Jest, JQuery

**Databases:** MySQL, PostgreSQL, SQLite, MongoDB, Redis

**Tools:** Git, GitHub, Bash, Linux, Notion, LaTeX, Docker, DigitalOcean, Caddy, Apache2, Nginx

## INTERESTS

Cycling, Traveling, Urban Exploration, Tennis, Kayaking